



Developing a computer-based tutor utilizing Generative Artificial Intelligence (GAI) and Retrieval-Augmented Generation (RAG)

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Abstract

This study investigates the development and evaluation of a Retrieval-Augmented Generation (RAG)-based statistics tutor designed to assist students with quantitative analysis methods. The RAG approach was employed to address the well-documented issue of hallucination in Large Language Models (LLMs). A computer tutor was developed that utilizes ChatGPT to understand student questions in natural language and retrieves relevant information from validated course notes to generate responses. The tutor was evaluated using twenty questions curated from the course notes. Two instructors who taught the course using the course notes assessed the quality of the tutor's response using a pre-defined rubric. The evaluation results indicate that the RAG-based tutor was able to provide correct and relevant answers to all twenty questions and to generate R code snippets upon request. This study highlights the potential of RAG-based tutors to provide accurate and personalized learning experiences for students while mitigating the risk of providing fabricated information often associated with LLMs.

Keywords Computer-based tutor · Generative Artificial Intelligence (GAI) · Retrieval-Augmented Generation (RAG) · Statistics education

1 Introduction

Artificial Intelligence (AI) has been heralded as a transformative force in education, especially for its potential to provide personalized learning experiences tailored to the needs of individual students (Holmes et al., 2019; Luckin et al., 2016). Traditional AI applications, such as Intelligent Tutoring System (ITS), have shown

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progress in adapting instructional content, pacing and feedback to individual learner (VanLehn, 2011; Kulik & Fletcher, 2016). These applications use computational algorithms to assess the performance of learners and provide targeted interventions and resources that can improve learning outcomes (Pane et al., 2014).

As the field of AI continues to evolve, the advent of Generative AI (GAI), such as ChatGPT, has further enhanced the feasibility of providing truly personalized education. GAI is built on Large Language Models (LLMs) trained on vast amounts of text data available on the Internet, enabling them to generate human-like text based on the input they receive. The underlying LLMs are fine-tuned through advanced deep-learning-based natural language processing techniques to improve performance in specific domains or tasks. GAI can generate contents that are dynamically tailored to each student's need, allowing for more nuanced and adaptive approaches to providing personalized learning experiences, where learning materials are created in response to student interactions. For instance, GAI can be employed to develop interactive tutorials that cater to students' specific questions and pace of learning, helping them to walk through complex concepts or problem-solving steps. This approach can potentially improve student learning by providing more personalized and adaptive learning experience.

As noted above, education researchers have been developing ITS to provide personalized and adaptive learning experience. Although ITS and GAI-based tutoring system share the common goal of providing customized learning experience to students, there exist a significant difference. ITS, such as Carnegie Mellon University's Cognitive Tutor (Koedinger et al., 1997), requires a rule-based cognitive model of target knowledge (e.g., how to add two fractions with different denominators) to understand student learning patterns and deliver tailored instructional supports. As a result, ITS typically focuses on structured subjects like mathematics and it may not be applicable to areas in which it is not easy to extract a set of explicit rules capturing important concepts and their relationships. On the other hand, GAI-based tutoring systems leverage natural language processing to engage in dynamic, conversation-based interactions with students, offering assistance across a much wider array of knowledge domains.

However, GAI-based tutoring systems have its own weaknesses. One significant weakness of GAI-based tutoring systems is tied to its knowledge cutoff date. The underlying GAI's training data only includes information available up to a specific date. Therefore, this temporal limitation can result in outdated or inaccurate responses from GAI-based tutoring systems. A more serious issue is that since the underlying GAI model "generates" its responses from language patterns learned from vast training datasets, it is susceptible to producing contents that, while sound plausible, might be factually incorrect or entirely fabricated, which is often called "hallucination" (Farquhar et al., 2024; Weise & Metz, 2023; Xu et al., 2024). In educational settings, where the accuracy of information is critical, the risk of hallucination is concerning.

In order to address the issue of hallucination in GAI, researchers have developed several approaches. First, since LLMs that power GAI tools acquire most of their knowledge during pretraining (Zhou et al., 2024), the hallucination problem can be mitigated by excluding unreliable, defective and noisy data that cannot be verified

while pretraining the model (Penedo et al., 2023; Touvron et al., 2023). Similarly, hallucinations can be reduced during the fine-tuning stage by using carefully curated prompts and responses (Zhou et al., 2024). Another approach to mitigating the hallucination problem is to employ reinforcement learning to distinguish desirable outputs from undesirable ones during the model building process (Lightman et al., 2023). Although these approaches have proven effective in mitigating hallucinations in LLMs, they require modifications to the internal structure of the LLM model (e.g., weights and biases of the neural network model), making it accessible only to computer scientists capable of building natural language models using deep learning.

Another approach to addressing these issues is Retrieval-Augmented Generation (RAG) (Lewis et al., 2020). Unlike conventional GAI models, which rely solely on learned patterns during training to generate responses, RAG integrates a retrieval mechanism to pull relevant information from predefined, validated sources. RAG addresses the problem of outdated knowledge by allowing the underlying LLM to incorporate up-to-date information, mitigating the impact of a static knowledge base. Moreover, RAG helps alleviate the hallucination problem by generating responses in the context of information retrieved from a validated knowledge base, enabling the model to verify its output (Extance, 2023). While fine-tuning updates the internal parameters of an LLM on a specific dataset to adapt the model to a particular task or domain, RAG tries to improve the LLM's performance by incorporating external knowledge sources without modifying the model's internal parameters (Balaguer et al., 2024). Since RAG does not involve changing the LLM's parameters, it requires much less computational effort and is much more accessible to those without a background in natural language processing and deep learning.

This exploratory study aims to answer the following research question: How effective is a GAI-based tutor employing RAG in providing answers to questions in quantitative research methods? To address this research question, a GAI-based statistics tutor was first developed using ChatGPT API (OpenAI, 2024a), LlamaIndex (LlamaIndex, 2024) and Streamlit (Streamlit, 2024). The quality of the tutor's explanations to free-form questions was then evaluated in terms of whether they included incorrect or excessive information. The rest of the paper is organized as follows. The [literature review](#) section summarizes previous studies on GenAI in education, hallucinations in GAI and RAG. The [methods](#) section explains what RAG is, how it was used in developing a computer tutor that can answer questions about statistical tests, and how the responses of this tutor were evaluated. The result section summarizes the correctness and excessiveness of the tutor's responses to evaluation questions. Finally, the paper concludes with discussions of implications and limitations of the study.

2 Literature review

2.1 GAI in education

Since the introduction of ChatGPT 3.5, education researchers' interest in GAI has surged as they observed its potential to revolutionize how we teach and learn new

knowledge. Despite this growing enthusiasm, research on GAI in education is still in its infancy. Most studies are largely conceptual or theoretical, focusing on how GAI might be used or discussing the pros and cons of its application in various educational contexts. For example, Lim et al. (2023) described the paradoxes of GAI, such as being both a friend and foe, capable yet dependent, accessible yet restrictive, and popular and banned, and explained the implications of incorporating GAI in education. Lodge et al. (2023) compared GenAI tools to calculators in terms of whether the tool is used individually or collaboratively, and whether the tool offloads or extends the cognition of learners. Tan et al. (2023) conceptualized how GenAI can be used to support collaborative learning. They articulated challenges and approaches for the sustainable, reliable, ethical and responsible use of GenAI in collaborative learning environment from the perspective of students, teachers, administrators and AI developers. Salah et al. (2023) examined the benefits and limitations of using ChatGPT in social psychology research, offering recommendations for its practical and responsible use. Hashmi and Bal (2024) explored the relationship between transparency and responsibility in the use of GenAI. They also discussed how higher education institutions could engage with GenAI to better prepare students for a future society where the use of GenAI would be more prevalent. Alasadi and Baiz (2023) discussed the impact of implementing AI in chemistry education in terms of advancing educational goals, ethical considerations, professional activities, equity and accessibility.

Another popular line of research explores the perceptions of students, teachers and administrators regarding the use of GAI in education. Chen et al. (2023) interviewed 10 high school students who used ChatGPT to examine how it supported their knowledge building efforts. Lee et al. (2024) surveyed 30 teaching staff and interviewed 8 of them regarding the impact of AI on higher education. Their thematic analysis revealed a lack of homogeneous sentiment about using AI, significant ambiguity regarding best practices for utilizing advanced technology tools, and concerns around academic integrity. Chan and Hu (2023) investigated students' perception of GenAI using a survey administered to 339 undergraduate and graduate students. They found that students generally had a positive attitude towards GenAI in teaching and learning, but also raised concerns about accuracy, privacy, ethical issues and the impact on personal development, career prospects and societal values. Kaplan-Rakowski and Grotewold (2023) explored teachers' perspectives on GAI and its potential implementation in education, focusing on its use, integration, potential and concerns. They found that teachers in general expressed positive attitudes towards GAI regardless of their teaching style. Moorhouse et al. (2023) examined how the world's 50 top-ranking higher education institutions developed or modified their assessment guidelines in response to emerging GAI. Smolansky et al. (2023) investigated the perspectives of educators and students on the impact of GenAI on assessments in higher education. They found that educators prefer adapted assessment incorporating GenAI and encouraging critical thinking, while students' responses were mixed due to concerns about a loss of creativity. While these discussions are crucial for understanding the broader implications of incorporating GAI in education, there is very little research that reports on the actual outcomes of utilizing GAI in educational contexts.

2.2 Hallucinations in generative AI

Generative Artificial Intelligence (GAI), while exhibiting remarkable capabilities, is known to produce responses containing false or misleading information (Ji et al., 2023). This phenomenon is commonly referred as hallucination, and it has been an important focus of research due to its implications in credibility of GAI-generated outputs. Much of research evaluating the credibility of text generated by GAI has centered on assessing the correctness of bibliographies generated by Large Language Models (LLMs). Sharun et al. (2023) utilized ChatGPT to compose 20 essays on stem cell research. The examination of 86 unique journal references cited in these essays revealed that 13 (15.12%) were fabricated, and 8 (9.30%) were erroneous references. Similarly, McGowan et al. (2023) focused on the accuracy of references generated by ChatGPT and Bard in a literature review on Psychiatry research. Out of the 35 references examined, only two were found to be real, with the majority were patchworks of citations from existing papers, raising concerns about models' ability to produce reliable academic content. Zuccon et al. (2023) investigated whether ChatGPT can provide evidence to supports its answers. Despite yielding correct or partially correct answers 51% of the time, the references suggested by ChatGPT existed only 14% of the time. Furthermore, even existing references often failed to support the claims attributed by ChatGPT. Walters and Wilder (2023) produced short literature reviews on various topics using ChatGPT 3.5 and ChatGPT 4. When they analyzed 636 bibliographic citations included in the literature review, they found that 55% of ChatGPT 3.5 citations and 18% of ChatCPT 4 citations were fabricated. Additionally, a significant portion of non-fabricated citations contained substantive errors. Maynez et al. (2020) conducted a large-scale human evaluation of neural abstractive summarization systems, including Transformer-based Generated Pre-Trained (GPT) language models. Across various systems, including GPT models, substantial amounts of hallucinated content were identified in generative summaries.

While most of previous studies investigating hallucination in AGI have focused on the bibliographies of GAI-generated text, there is a growing number of studies examining different sources of text. For instance, Rawte et al. (2023) used a data set of New York Times news event tweets to investigate how linguistic factors, such as readability (Flesch, 1948), formality (Heylighen & Dewaele, 1999) and concreteness (Brysbaert et al., 2014), influence the occurrence of hallucinations. Their analysis showed that more complex, less formal, and more abstract prompts resulted in more hallucinations. Varshney et al. (2023) examined the occurrence of hallucinations in articles generated by ChatGPT 3.5 across various categories (e.g., sports, music, politics, film and TV, history, services, science, art, literature and others). When they manually checked the correctness of the generated text using Web searches, they found that about half of the generated text contained incorrect information. Ladhak et al. (2023) used a Wikipedia Name-Nationality dataset ("Lists of people by nationality", 2024) to investigate the name-nationality hallucinations in LLM-generated summary text. They found that the hallucination rate was significantly higher for Asian nationalities (about 33%) compared to those from countries in America (less than 5%). McIntosh et al. (2024) evaluated the responses from several

LLMs, including ChatGPT 3.5, ChatGPT 4, Bard, Perplexity AI and TruthGPT, to questions in 7 categories (religious disputes, political disagreements, ethical dilemmas, pseudoscience, social norm expectations, health misinformation and brain teasers). They found that topics lacking a clear scientific consensus and the brain teasers were more susceptible to hallucinations.

2.3 Addressing hallucinations with Retrieval-Augmented Generation (RAG)

Recently, researchers have begun exploring how RAG can mitigate GAI's hallucination problem. Zakka et al. (2024) demonstrated that an LLM augmented with retrieval capabilities outperformed standard LLMs in the medical field, as evaluated by experts across multiple specialties. Similarly, Ge et al. (2023) showcased the successful integration of protected health information into an LLM tailored for liver disease inquiries. Lála et al. (2023) highlighted the advantages of RAG, showing improved performance in answering questions over scientific literature compared to traditional LLMs. Saba et al. (2024) emphasized the benefits of RAG, including minimal training requirements, diversity in responses, and avoidance of hallucination issues common in LLMs. Shuster et al. (2021) supports the potential of RAG in mitigating hallucination problems in open-domain question answering, while Lewis et al. (2020) presents a RAG model that outperforms other NLP models on knowledge-intensive tasks, leveraging both pre-trained seq2seq models and dense vector indices. Finally, Wang et al. (2024) demonstrated the effectiveness of RAG in synthesizing information from multiple sources for personalized response generation. Overall, these studies collectively underscore the value of RAG in enhancing the capabilities and reliabilities of language models across diverse applications.

3 Methods

3.1 RAG-Based statistics Tutor

This study aims to develop and evaluate a computer tutor that communicate with students in natural language when they need help on quantitative research methods. To enable the tutor to understand students' questions, the tutor developed in this study utilized ChatGPT API (OpenAI, 2024a). Additionally, to prevent the tutor from providing incorrect information, Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) was employed in developing the tutor.

Figure 1 depicts how the computer tutor is utilizing RAG in answering a question. First, the text in the course notes, which covers t-test and ANOVA, are broken into smaller chunks to overcome ChatGPT's limitation on the size of contextual information (Chunks). These smaller chunks are then converted into a sequence of numeric values called embedding to help the tutor locate most relevant chunks to a user's question (Embedded Chunks). The statistics tutor developed in this study utilized OpenAI's "text-embedding-ada-003" embedding model which converts a given text into a series of 3,071 numerical values (OpenAI, 2024b). Since text

embeddings can capture the semantic meaning of words and phrases in documents, it allows us to find relevant information based on “semantic similarity” rather than exact word matching (Jurafsky & Martin, 2024). The embedded smaller chunks are stored in a vector database for searching. The statistics tutor developed in this study used Chroma (Chroma, 2024), an open-source vector database, to store embedded text chunks and perform semantic search. When a user asks a question, the question is converted into a sequence of numeric values using the same embedding model (Embedded Question). The embedded user question is used to select the most relevant chunks from the Chroma vector database (Retrieve). The retrieved chunks and the user’s question are then sent to ChatGPT 3.5 (Augment) with an instruction that the retrieved chunks should be used as context for answering the user’s question (Generate). The statistics tutor developed in this study utilized an open-source Python library called LlamaIndex (LlamaIndex, 2024) to implement a RAG approach. Specifically, it used LlamaIndex to generate a question based on the relevant chunks retrieved from the Chroma vector database and the history of tutor-user conversations. It then requested ChatGPT 3.5 to answer the question based on the relevant chunks and conversation history. Since ChatGPT generates its response from the provided the context (relevant chunks and conversation history), it is less likely to produce a fabricated answer. Also, when the tutor asked ChatGPT-3.5 to answer questions, it set the “temperature” parameter of ChatGPT, which controls the level of randomness in ChatGPT’s responses, to zero to make its responses as deterministic as possible. As a result, the tutor received very similar answers, though not exactly same, since the response is still “generated” by LLM, when the same question was asked repeatedly.

Streamlit (Streamlit, 2024), an open source Python library that makes it easy to create Web applications for machine learning and data science, was used to

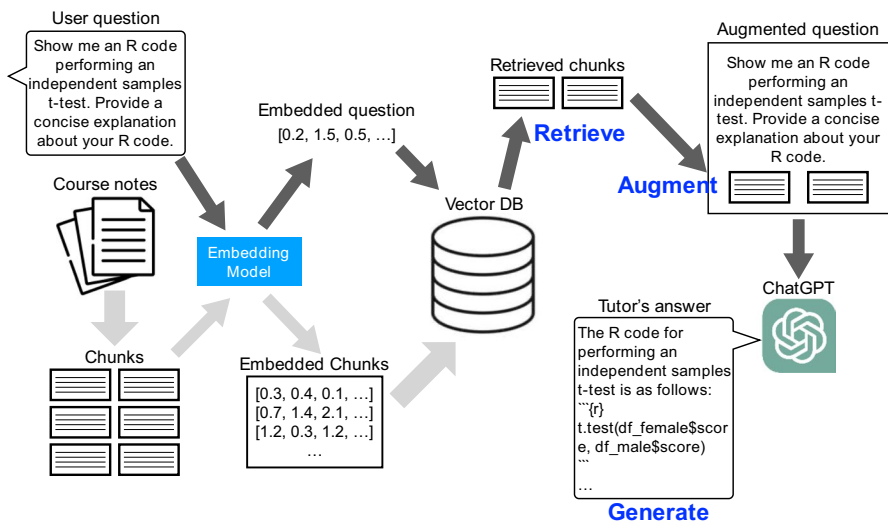


Fig. 1 Retrieve-Augment Generation (RAG) process for answering questions

deploy the tutor as an interactive Web application. Figure 2 shows a screen-capture image of the RAG-based statistics tutor developed in this study.

3.2 Evaluating RAG-based statistics tutor

Evaluating an LLM differs from traditional training/test set approaches in machine learning. While the latter focuses on quantifiable metrics such as accuracy, precision and recall derived from labeled test data, evaluating an LLM extends beyond such performance metrics because there exist multiple ways to convey the same idea in natural language. As a result, LLM evaluation often requires human judgement to assess the qualities of LLM's responses, making it inherently subjective.

In Natural Language Generation (NLG) and Machine Translation (MT), BLEU (BiLingual Evaluation Understudy) (Papineni et al., 2002) and ROGUE (Recall-Oriented Understudy for Gisting Evaluation) (Lin, 2004) have been used as a performance metric. These approaches measure the number of matching “n-grams” between the reference and the generated text. However, Liu et al. (2023) showed that these metrics have low correlations with human judgements. In this study, therefore, the two instructors of a quantitative research methods course offered to graduate students majoring in educational technology assessed the quality of the tutor's responses. One instructor who created the course note based on open educational resources, such as Quantitative Methods using R (Parajuli, 2023) and Applied Statistics with R (Dalpiaz, 2022), had an extensive research experience over fifteen years in quantitative data analysis and educational data mining. The other instructor also has a strong background in quantitative research, having conducted her research using statistics and learning analytics for over five years. The instructor who created the course notes prepared 4 easy, 10 medium, and 6 hard questions from them (see Table 1). Some questions require a conceptual understanding of statistics (e.g., question 11) while others are asking for R commands necessary for specific statistical analysis methods (e.g., question 13). The difficulty of each question was determined by the number of sub-tasks and the complexity of the concepts involved. For instance, Q11 is considered more difficult than Q2, even though both are conceptual questions, because the concept addressed in Q11 (ANOVA) is more advanced than the one in Q2 (t-test). Also, a question was considered more difficult if it involved more sub-tasks. For example, Q18 is considered more difficult than Q16, even though both involve the same statistical test (ANOVA), because the answer to Q18 includes three parts, while the answer to Q16 involves only one. These 20 questions were input into the RAG-based tutor to simulate questions from students taking the course, in order to generate the corresponding answers.

The instructors then assessed the correctness and quality of the tutor's responses using a rubric outlined in Table 2. Note that the rubric evaluates not only accuracy and relevance, but also the presence of excessive information, indicating that the ideal answer should contain just the right amount of correct and relevant information for the question at hand.

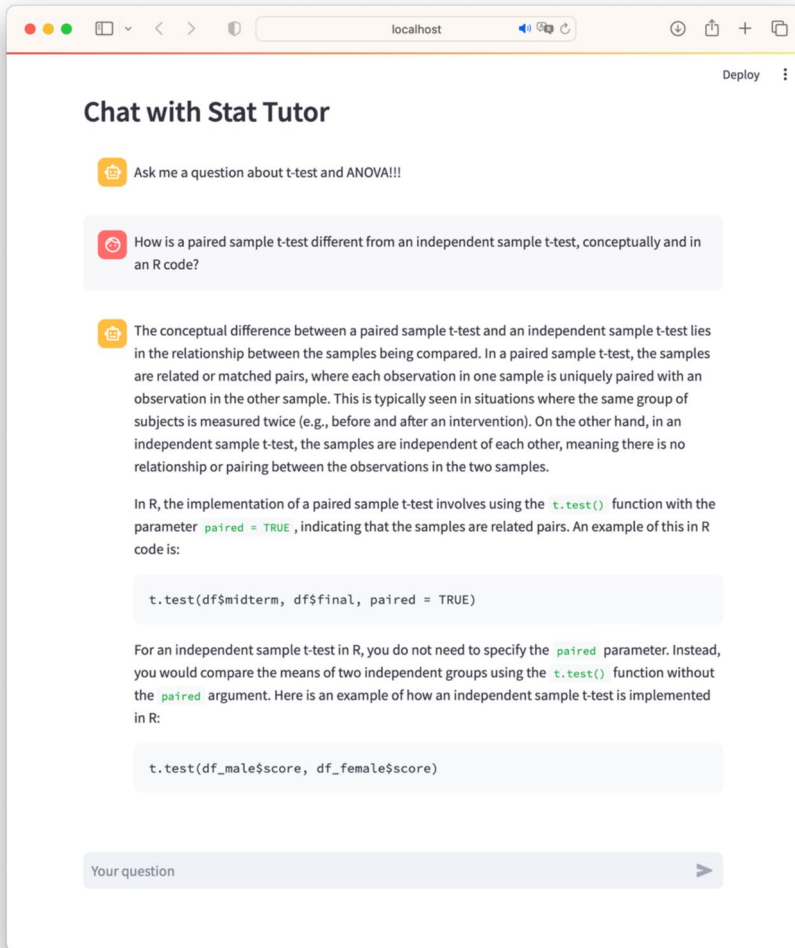


Fig. 2 RAG-based tutor developed in this study. It can understand questions in natural language and can provide answers, including explanations and R source code

4 Results

Table 3 shows the tutor's responses. Overall, the tutor successively provided appropriate answers to all questions. The length of tutor's response varied, depending on the complexity of the question. Upon request, the tutor was able to provide working R code that students could use.

The length of tutor's response ranged from 13 to 336 words. Unsurprisingly, the RAG-based tutor provided a more detailed and longer response when the

Table 1 Twenty questions used to evaluate the RAG-based tutor and the difficulty of problems

Question	Difficulty
Q1. Show me an R code computing a mean of values in the column in a data frame df.	Easy
Q2. What kinds of statistical test can we use when comparing mean values from two groups?	Easy
Q3. What kinds of t-test exist?	Medium
Q4. What kinds of descriptive statistics we often use in performing a t-test?	Easy
Q5. Show me an R code visualizing data for a t-test.	Medium
Q6. Show me an R code performing an independent samples t-test. Provide a concise explanation about your R code.	Medium
Q7. How is a paired samples t-test different from an independent samples t-test, conceptually and in an R code?	Hard
Q8. How can we assess the magnitude of difference in means from two groups? Include an R code in your response.	Hard
Q9. How do we know if our data is appropriate for t-test? Show me an R code.	Medium
Q10. Explain how to interpret an outcome of a t-test in R. Include an R code in your response.	Hard
Q11. How is ANOVA different, compared to t-test?	Easy
Q12. Show me an R code creating a box and whisker plot using tidyverse, and explain function arguments.	Medium
Q13. Show me an R code computing descriptive statistics, mean, median and standard deviation, on groups for ANOVA	Medium
Q14. What is an alternative hypothesis in ANOVA?	Medium
Q15. How can we perform an ANOVA using R? Explain important parts in the ANOVA test result in R.	Hard
Q16. What is an effect size statistic often used in ANOVA?	Medium
Q17. Show an R code computing an effect size for ANOVA, and explain how to interpret it.	Hard
Q18. How do we know where the difference identified in ANOVA comes from? What kinds of statistical test can we use to answer that question? Show me an R code doing that test.	Hard
Q19. What kind of information do we need to include when we report the findings from an ANOVA test?	Medium
Q20. What does it mean to have a p-value less than 0.05 in ANOVA?	Medium

Table 2 The rubric for evaluating the RAG-based tutor's response

Description	Score
The tutor's response is incorrect.	1
The tutor's response is insufficient and contains incorrect information.	2
The tutor's response includes all necessary information from the course notes, but also contains incorrect information.	3
The tutor's response includes only correct information from the course notes, but is insufficient.	4
The tutor's response includes all necessary information from the course notes, but also includes additional, unnecessary information that may not be directly related to the question.	5
The tutor's response includes sufficient information for the question, and all information in the response is related to the question. However, the response may be too lengthy because it includes excessive information.	6
The tutor's response includes sufficient information for the question, and all the information in the response is related to the question.	7

question was more difficult. Table 4 shows descriptive statistics of the number of words in the tutor’s response.

Table 5 presents the assessment of two instructors who taught different sections of the same quantitative research methods course, using the same course notes that served as the source documents for the RAG-based statistics tutor developed in this study. For most questions, both evaluators agreed that the RAG-based tutor delivered responses that were both sufficient and accurate (score=7). Although the tutor’s responses to two questions (Q2 and Q5) were found to be insufficient, they did not contain any incorrect information. This result suggests that the RAG approach effectively addresses the issue of hallucination commonly associated with LLMs. The responses to questions Q8, Q10 and Q12 were noted by one or both instructors to contain excessive information, which could potentially confuse students.

The Quadratic Weight Kappa (QWK), a measure of agreement between two raters classifying items into mutually exclusive categories (Cohen, 1968), was calculated to assess inter-rater reliability. QWK extends Cohen’s Kappa by more heavily penalizing disagreements proportionally to the squared distance between categories.

Table 3 Stat tutor’s response to evaluation questions

Question	Tutor Response
Q1	<pre>```{r} mean(df\$column_name) ```</pre> Replace `column_name` with the name of the specific column in the data frame `df` for which you want to compute the mean.
Q2	We can use an independent samples <i>t</i> -test when comparing mean values from two groups.
Q3	We can use <i>t</i> -tests to compare mean values from two groups. There are two different types of <i>t</i> -tests: independent samples <i>t</i> -test and paired-samples <i>t</i> -test. The independent samples <i>t</i> -test is used when the two groups are not related, while the paired-samples <i>t</i> -test is used when the two groups are related.
Q4	Descriptive statistics commonly used when performing a <i>t</i> -test include measures such as the mean and standard deviation. These statistics provide information about the central tendency and variability of the data, which are important for understanding the distribution of the variables being compared in the <i>t</i> -test.
Q5	The R code for visualizing data for a <i>t</i> -test is to use the `ggplot` function from the `ggplot2` package. You can create a histogram of the data using the `geom_histogram` function, and customize the plot using additional functions like `labs` to label the axes.
Q6	The R code for performing an independent samples <i>t</i> -test is as follows: <pre>```{r} t.test(df_female\$score, df_male\$score) ```</pre> This code uses the `t.test()` function in R to perform the <i>t</i>-test. The function takes two numeric vectors as input, which in this case are the math test scores of female students (`df_female\$score`) and male students (`df_male\$score`). The <i>t</i>-test compares the means of these two groups and provides several outputs, including the test statistic value, the degrees of freedom, the <i>p</i>-value, and the alternative hypothesis. The <i>p</i>-value is used to determine the statistical significance of the difference in means between the two groups.

Table 3 (continued)

Question	Tutor Response
Q7	A paired sample <i>t</i>-test is used when the observations in the two groups are related or paired in some way. This means that each observation in one group is matched or paired with a corresponding observation in the other group. The purpose of the test is to determine if there is a significant difference between the means of the paired observations. On the other hand, an independent sample <i>t</i>-test is used when the observations in the two groups are independent and not related to each other. The purpose of this test is to determine if there is a significant difference between the means of the two independent groups. In terms of R code, the main difference lies in the argument passed to the <i>t</i>-test function. For a paired sample <i>t</i>-test, the argument "paired = TRUE" is used to indicate that the observations are paired. For an independent sample <i>t</i>-test, this argument is not needed as the default assumption is that the observations are independent. Here is an example of R code for a paired sample <i>t</i>-test: <pre>```{r} t.test(group1, group2, paired = TRUE) ```</pre> And here is an example of R code for an independent sample <i>t</i>-test: <pre>```{r} t.test(group1, group2) ```</pre>
Q8	To assess the magnitude of difference in means from two groups in R, we can compute an effect size. One commonly used effect size measure is Cohen's <i>d</i>. Here is an R code to compute Cohen's <i>d</i>: <pre>```{r} # Assuming we have two groups, group1 and group2 mean1 <- mean(group1) mean2 <- mean(group2) sd1 <- sd(group1) sd2 <- sd(group2) # Compute Cohen's d cohens_d <- (mean1 - mean2) / sqrt((sd1^2 + sd2^2) / 2) cohens_d ```</pre> In this code, we first calculate the means (`mean1` and `mean2`) and standard deviations (`sd1` and `sd2`) of the two groups. Then, we compute Cohen's <i>d</i> by taking the difference between the means and dividing it by the pooled standard deviation. Cohen's <i>d</i> provides a standardized measure of the difference between the means, where a larger value indicates a larger effect size.

Table 3 (continued)

Question	Tutor Response
Q9	We can determine if our data is appropriate for a t-test by checking if the data is normally distributed. One way to assess the normality of the data is by creating a histogram and visually inspecting the shape of the distribution. Another way is to perform a Shapiro-Wilk normality test, which provides a statistical measure of the normality of the data. To assess the appropriateness of the data using R, we can use the <code>`shapiro.test()`</code> function. Here is an example code: <pre> ```{r} shapiro.test(data) ``` </pre> Replace "data" with the name of your dataset. The <code>`shapiro.test()`</code> function will return the p-value of the Shapiro-Wilk test. If the p-value is greater than the predetermined significance level (usually 0.05), we can conclude that the data is normally distributed and appropriate for a t-test. If the p-value is less than the significance level, we would conclude that the data is not normally distributed and may not be appropriate for a t-test.
Q10	The outcome of a t-test in R can be interpreted by examining the p-value. In R, we can use the <code>`t.test()`</code> function to perform a t-test and obtain the p-value. The p-value represents the probability of observing the data, or more extreme data, if the null hypothesis is true. If the p-value is less than a pre-determined significance level (usually 0.05), we reject the null hypothesis and conclude that there is evidence of a significant difference between the groups being compared. On the other hand, if the p-value is greater than the significance level, we fail to reject the null hypothesis and conclude that there is not enough evidence to suggest a significant difference between the groups. Here is an example of how to interpret the outcome of a t-test in R: <pre> ```{r} # Example t-test group1 <- c(1, 2, 3, 4, 5) group2 <- c(6, 7, 8, 9, 10) result <- t.test(group1, group2) # Interpretation if(result\$p.value < 0.05) { print("There is a significant difference between the groups.") } else { print("There is not enough evidence to suggest a significant difference between the groups.") } ``` </pre> In this example, if the p-value is less than 0.05, we would interpret the result as "There is a significant difference between the groups." If the p-value is greater than or equal to 0.05, we would interpret the result as "There is not enough evidence to suggest a significant difference between the groups."
Q11	ANOVA is different from a t-test in that it allows for the comparison of means across multiple groups, whereas a t-test only compares means between two groups. ANOVA is used when there are more than two groups to compare, while a t-test is used when there are only two groups. ANOVA computes an F statistic and tests whether there is a significant difference in means among the groups, while a t-test computes a t statistic and tests whether there is a significant difference in means between the two groups.

Table 3 (continued)

Question	Tutor Response
Q12	To create a box and whisker plot using tidyverse in R, you can use the <code>`geom_boxplot()`</code> function. This function is part of the <code>ggplot2</code> package, which is included in the tidyverse. The <code>`geom_boxplot()`</code> function takes several arguments to customize the appearance of the plot. Here are the main arguments: - <code>`data`</code>: This argument specifies the data frame that contains the variables you want to plot. In the given context, the <code>`df2`</code> data frame is used, which contains the <code>`gear`</code> and <code>`mpg`</code> variables. - <code>`mapping`</code>: This argument specifies how the variables in the data frame should be mapped to the aesthetics of the plot. In the given context, <code>`x = gear`</code> and <code>`y = mpg`</code> are specified to map the <code>`gear`</code> variable to the x-axis and the <code>`mpg`</code> variable to the y-axis. - <code>`fill`</code>: This argument specifies the fill color of the boxes in the plot. By default, the boxes are filled with a solid color. You can specify a specific color or use a variable from the data frame to map the fill color. - <code>`color`</code>: This argument specifies the color of the outlines of the boxes and whiskers. By default, the outlines are black. You can specify a specific color or use a variable from the data frame to map the color. - <code>`size`</code>: This argument specifies the size of the outlines of the boxes and whiskers. By default, the size is 0.5. You can specify a different size value to make the outlines thicker or thinner. - <code>`alpha`</code>: This argument specifies the transparency of the boxes and whiskers. By default, the transparency is 1 (fully opaque). You can specify a value between 0 and 1 to make the boxes and whiskers more transparent. These are just a few of the arguments that can be used with <code>`geom_boxplot()`</code>. There are many more options available to customize the appearance of the plot, such as adding labels, changing axis titles, adjusting the position of the boxes, and more. You can refer to the <code>ggplot2</code> documentation for more details on the available options.
Q13	To compute descriptive statistics (mean, median, and standard deviation) on groups for ANOVA using R, you can use the <code>`group_by()`</code> and <code>`summarize()`</code> functions from the <code>`dplyr`</code> package. First, use the <code>`group_by()`</code> function to specify the grouping variable. Then, use the <code>`summarize()`</code> function to calculate the desired statistics for each group. For example, to compute the mean, median, and standard deviation of a variable <code>`x`</code> for each group defined by a factor variable <code>`group`</code>, you can use the following code: <pre> {r} df %>% group_by(group) %>% summarize(mean = mean(x), median = median(x), sd = sd(x)) </pre> Replace <code>`df`</code> with the name of your data frame, <code>`group`</code> with the name of your factor variable, and <code>`x`</code> with the name of the variable for which you want to compute the statistics. The resulting data frame will contain the mean, median, and standard deviation for each group.
Q14	The alternative hypothesis in ANOVA is that the mean values in each group are not the same.

Table 3 (continued)

Question	Tutor Response
Q15	To perform an ANOVA test in R, you can use the <code>`aov()'</code> function. The formula in the <code>`aov()'</code> function specifies the independent variable and the dependent variable. You also need to provide the name of the R object containing the data. The result of the ANOVA test is saved in a new R object, which can be evaluated later. The important parts in the ANOVA test result in R are the F statistic, the degrees of freedom, and the p-value. The F statistic measures the ratio of the between-group variability to the within-group variability. The degrees of freedom represent the number of independent pieces of information used to estimate the population parameters. The p-value indicates the probability of observing the F statistic or a more extreme value if the null hypothesis is true. If the p-value is smaller than a pre-determined significance level (usually 0.05), the null hypothesis is rejected, indicating that there is a statistically significant difference in the mean values among the groups.
Q16	The effect size statistic often used in ANOVA is ω^2 (omega squared).
Q17	To compute an effect size for ANOVA using R, you can use the <code>`effectsize()'</code> function from the <code>`effectsize'</code> package. Specifically, you can provide "omega" as the type of effect size. This will compute the effect size measure ω^2, which represents the proportion of variability in the dependent variable explained by the independent variable. To interpret the effect size, you can use the following guidelines: - $\omega^2 < 0.01$: Very small effect size - $0.01 \leq \omega^2 < 0.06$: Small effect size - $0.06 \leq \omega^2 < 0.14$: Medium effect size - $\omega^2 \geq 0.14$: Large effect size These guidelines provide an indication of the magnitude of the difference between groups. A larger effect size indicates a stronger relationship between the independent and dependent variables.
Q18	To determine where the difference identified in ANOVA comes from, we can perform a post-hoc multiple comparison test. One commonly used test for this purpose is the Tukey's Honestly Significant Difference (Tukey's HSD) test. In R, we can use the <code>`TukeyHSD()'</code> function to perform the Tukey's HSD test. This function takes the ANOVA result object as input and provides the pairwise comparisons between the groups along with the adjusted p-values. Here is an example of how to perform the Tukey's HSD test in R: <pre> '''{r} tukey_result <- TukeyHSD(anova_result) tukey_result ''' </pre> The <code>`tukey_result'</code> object will provide the pairwise comparisons between the groups, along with the difference in means and the adjusted <math>p</math>-values. This will help us identify which specific groups have significantly different mean values.</p> <p>Note that the <code>`anova_result'</code> object is the result of the ANOVA test performed earlier in the code.
Q19	When reporting the findings from an ANOVA test, we need to include the test statistic (F), the degrees of freedom (df), the p -value, and the effect size measure (ω^2).
Q20	Having a p -value less than 0.05 in ANOVA means that the observed difference in the mean values among groups is statistically significant. In other words, it is unlikely to observe such a large difference in the mean values if the null hypothesis (no difference in mean values) is true. Therefore, we reject the null hypothesis and conclude that there is a significant difference in the mean values among the groups.

Table 4 Descriptive statistics of number of words in the tutor's response

Difficulty	Mean	SD	Median	IQR
Easy	45.25	34.49	36.50	35.25
Medium	99.00	98.30	64.50	98.75
Hard	174.33	48.23	159.50	58.00

$$QWK = 1 - \frac{\sum O_w(i,j)}{\sum E_w(i,j)}$$

Where, $O_w(i,j)$ and $E_w(i,j)$ are the observed agreement and the expected agreement by chance, weighted by the degree of disagreement with $W(i,j) = \frac{(i-j)^2}{(N-1)^2}$. The QKT value indicates that the two instructors had a very strong agreement regarding the quality of tutor's responses ($QWK=0.884$, $p<.0001$).

5 Discussion

This study explored the development and evaluation of a Retrieval-Augmented Generation (RAG)-based statistics tutor aimed at assisting students in learning two statistical analysis methods: t-test and ANOVA. The results show the RAG approach's efficacy in reducing the hallucination problem prevalent in Large Language Models (LLMs), while also highlighting the potential of RAG-based tutors to enhance student learning.

A significant issue with LLMs is their tendency to produce fabricated or misleading content, a problem referred to as hallucination (Farquhar et al., 2024; Weise & Metz, 2023; Xu et al., 2024). This study demonstrated that a RAG approach could effectively tackle this issue. The RAG-based tutor consistently provided accurate responses to all evaluation questions, as verified by two human experts, without generating any factually incorrect information. This indicates that the RAG approach, by leveraging information retrieved from validated sources, successfully mitigated the hallucination problem and ensured the precision of its responses.

Furthermore, the RAG-based tutor proved to be a versatile educational tool. It adeptly addressed questions requiring a range of understanding, from fundamental statistical concepts to providing working R code for data analysis. Such flexibility can allow the tutor to meet diverse learning needs. Students grappling with specific concepts can turn to the tutor for a more comprehensive understanding, while those more advanced may use it to quickly generate R code snippets, thus facilitating their data analysis projects.

While the findings are encouraging, there is room for improvement. The variability in the length of the tutor's responses, depending on the complexity and nature of questions, was observed. Instructors evaluating the tutor's responses noted that some answers might include too much information, providing excessive details that would potentially overwhelm students. This issue is particularly relevant for novice learners who may struggle to deal with too much information that exceeds the scope of their questions or their current level of understanding. The cognitive

Table 5 Quality of tutor's responses to evaluation questions

Question	Score (Instructor 1)	Score (Instructor 2)
Q1	7	7
Q2	4	4
Q3	7	7
Q4	6	6
Q5	4	4
Q6	7	7
Q7	7	7
Q8	7	6
Q9	7	7
Q10	7	6
Q11	7	7
Q12	6	6
Q13	7	7
Q14	7	7
Q15	7	7
Q16	7	7
Q17	7	7
Q18	7	7
Q19	7	7
Q20	7	7

load imposed by overly detailed responses could hinder their understanding rather than support their learning (Sweller, 1988; Sweller et al., 2011; van Merriënboer et al., 2003). However, it is important to note that the RAG-based tutor can answer follow-up questions, allowing students to request more concise explanations if they find the tutor's response too long or complicated. This interactivity can help mitigate the issue by encouraging students to engage in further dialogues with the tutor, highlighting the importance of training students to effectively interact with the RAG-based tutor. Teaching students the strategies for managing the depth and breadth of information and how to request follow-up explanations will empower them to use the tutor more efficiently. This would not only improve their learning experience, but also help them develop the critical thinking skills necessary to identify relevant information and engage more effectively in self-regulated learning.

The RAG-based approach applied in this study is inherently flexible, capable of being adapted to various subjects, thereby providing a versatile framework for enriching student learning experience across diverse academic fields. It would be highly meaningful to conduct user studies evaluating student perceptions and the educational impact of interacting with the RAG-based tutor. Such studies could offer valuable insights into how students engage with the tutor's responses, revealing whether the tutor improves their engagement, motivation, and understanding of key concepts under study. Also, assessing the educational impact, such as improvements in learning outcomes, retention

of information, and problem-solving skills, would provide empirical evidence of the tutor's effectiveness. This line of research could also help identify areas for improvement in the development of GAI-based tutors, such as fine-tuning the level of detail in the tutor's responses or addressing any usability issues encountered by students.

Considering that the RAG-based tutor requires an OpenAI API subscription, its use is limited to educational institutions with sufficient resources, which could exacerbate the digital divide in our society. One way to address this issue is developing a RAG-based tutor using an open-source LLM such as Ollama (Ollama, 2024). This approach can democratize access to advanced educational tools like GAI-based learning environments, particularly for institutions with limited resources.

In this research, the evaluation of the tutor's responses was conducted by two human experts. While expert evaluation is considered the gold standard, it is time-consuming and challenging to scale. Recent research by Zheng et al. (2023) examined the potential of employing advanced LLMs like GPT-4 as evaluators of LLM-generated content. Their findings indicate that these sophisticated LLM evaluators can attain an accuracy rate exceeding 80%, which is comparable to the level of agreement typically observed among human judges. Comparing the assessments made by human experts in this study with those performed by advanced LLM judges could provide valuable insights, suggesting a more scalable solution for evaluating the effectiveness of the RAG-based tutor.

Finally, since hallucination tends to occur more frequently with topics lacking a clear scientific consensus (McIntosh et al., 2024) and with more complex prompts (Rawte et al., 2023), it is possible that hallucination may not be a serious issue for ChatGPT without RAG when responding to questions about performing t-test and AVNOA using R. Thus, it would be meaningful to compare the responses from the RAG-based tutor to those from a standard ChatGPT model that does not employ RAG.

6 Conclusion

This study demonstrated the potential of a RAG-based tutor to support students' learning in statistical concepts, such as t-test and ANOVA, by providing relevant responses without hallucination. The findings of this study highlight both the strengths and areas for improvement in RAG-based learning environments. While the tutor was able to provide correct answers, it sometimes produced excessive information. Addressing this challenge, along with evaluating student perceptions and the educational impact through user studies, will be essential for improving the tutor's effectiveness. Additionally, the use of an open-source LLM offers a promising solution for making GenAI-based educational technology more accessible to educational institutions with limited resources, helping to bridge the digital divide.

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Data availability The author confirms that all data analyzed in this study are included in this published article.

Declarations

Conflict of Interest None.

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